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NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 2 EXAMINATION 2017-2018

CE2005/CZ2005 – OPERATING SYSTEMS

Apr/May 2018 Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 8 pages.
2. Answer **ALL** questions.
3. This is a closed-book examination.
4. Questions **DO NOT** carry equal marks.

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1. (a) Indicate whether each of the following statements is true or false. Justify each answer.

(i) To ensure security, the monitor mode should be used for all program instructions.

(2 marks)

(ii) Round-Robin scheduling is not always appropriate for online, interactive systems.

(2 marks)

(iii) CPU scheduling is preemptive if it happens only when a new process is admitted into the system.

(2 marks)

(iv) Multi-programming has little benefit in multi-core systems.

(2 marks)

(v) Interrupt-driven I/O can cause much CPU overhead.

(2 marks)

Note: Question No. 1 continues on Page 2

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(b) Table Q1 shows some processes with their respective CPU-burst time and arrival time. P3 and P4 arrive at time 2 in the order of P3 then P4. The timer interrupt occurs every 3 units of time.

Table Q1

Process	Burst Time	Arrival Time
P1	10	0
P2	5	3
P3	2	2
P4	6	2

(i) Calculate the average turnaround time considering the following scheduling algorithms: Round-Robin (implemented using the timer interrupt), Non-preemptive Shortest Job First, and Shortest Remaining Time First.

(3 marks)

(ii) Design another scheduling algorithm which would have exactly the same performance in terms of average turnaround time as Non-preemptive Shortest Job First. The new algorithm should be different from those considered in Q1(b)(i).

(3 marks)

(c) The CPU utilization of a system is currently very low. Suggest and justify appropriate scheduling actions the operating system should carry out to improve CPU utilization.

(4 marks)

(d) A program has been written to do the following tasks sequentially: read a number from the keyboard, compute an arithmetic expression, and store the result into a shared memory location.

(i) Identify and justify all the mode switches in the above program.

(2 marks)

(ii) Two processes P1 and P2 have been created from this program, all at time 0, in the order of P1 then P2. List and explain all the possible context switches for P1 and P2, assume that First-Come First-Serve scheduling is used.

(3 marks)

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2. (a) Indicate whether each of the following statements is true or false. Justify each answer.

(i) If the semaphore wait() operation is not atomic, mutual exclusion is violated.

(2 marks)

(ii) A process runs quickly so it can cut the queue into the critical section all the time. This is a progress violation.

(2 marks)

(iii) Desktop operating systems like Windows never have deadlock issues.

(2 marks)

(iv) For semaphore implementation, blocking is not always better than busy waiting.

(2 marks)

(v) A system is not safe if there is at least one unsafe sequence of process executions.

(2 marks)

(b) A system has three processes (P1, P2 and P3) and three resource types (A, B, and C). The current Allocation and Need matrices are shown in Table Q2b.

Table Q2b

		Available
		A B C
		2 2 0
Process	**Allocation**	**Need**
	A B C	A B C
P1	1 2 1	2 1 4
P2	1 2 3	1 2 0
P3	1 0 2	3 1 1

(i) Justify if the system is currently safe or not.

(2 marks)

(ii) Justify if a request for (A=2, B=1, C=0) from P3 should be granted.

(2 marks)

Note: Question No. 2 continues on Page 4

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(c) Consider a producer-consumer synchronization problem which has multiple concurrent producers and consumers. Each of these processes requires exclusive access to a buffer which can hold 10 items at most. If the buffer is empty, the consumers should wait. Similarly, if the buffer is full, the producers should wait. The pseudo-code of an initial solution using semaphores is given in Table Q2c.

Table Q2c

Semaphores	Producer process	Consumer process

full, initialized to 0
 	
empty, initialized to 10
 	
mutex, initialized to 1
	item *p*;
	while (1) {
	wait(*mutex*);
	wait(*empty*);
	produce an item *p;*
	add *p* to *buffer;*
	signal(*full*);
	signal(*mutex*);
	}
		item *c*;
		while (1) {
		wait(*mutex*);
		wait(*full*);
		c = an item from b*uffer;*
		signal(*empty*);
		signal(*mutex*);
		consume item *c;*
		}

(i) Identify and explain <u>two</u> problems with the above solution.

(4 marks)

(ii) Blocking implementation is used for all semaphores. Considering the solution given in Table Q2c, what are the minimum and maximum values for *mutex* assuming that there are a total number of 10 processes in the system?

(2 marks)

(iii) Modify the pseudo-code in Table Q2c so that a consumer must consume all the available items once it can gain access to the buffer. Use additional semaphores, variables, statements, etc. if necessary. Your solution has to be correct and deadlock-free.

(5 marks)

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3. (a) Indicate whether the following statements about memory management are true or false. Justify your answers.

(i) If address binding is performed during execution time, addresses generated by CPU can be used directly to access physical memory.

(2 marks)

(ii) Memory allocation using paging does not have internal fragmentation problem.

(2 marks)

(iii) Consider a paging system with page table stored in memory. Using translation look-aside buffer (TLB) will always reduce effective memory access time.

(2 marks)

(iv) If segmentation is used for memory allocation, the logical address space of a process will remain contiguous in physical memory.

(2 marks)

(v) If its reference bit is set, a page needs not to be written back to the backing store when being replaced.

(2 marks)

(vi) Thrashing can be detected by evaluating the level of CPU utilization as compared to the level of multiprogramming.

(2 marks)

(b) Consider a demand paging system that uses a fixed allocation policy for frame allocation. Assume logical address has 12 bits, page size is 1,024 bytes, and there are 8 frames in the physical memory. Figure Q3b shows the page table of a process. A dash in an entry indicates the corresponding page is not in the memory. As shown in the figure, three frames are allocated to the process.

Page Number	Frame Number	Valid Bit	Time Loaded	Time Last Accessed
0	2	1	120	200
1	7	1	150	160
2	-	0	-	-
3	4	1	140	180

Figure Q3b

Note: Question No. 3 continues on Page 6

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(i) How many bits are there in physical address?

(2 marks)

(ii) What are the results of translating the following logical addresses (in decimal form) to the corresponding physical addresses (in decimal form):

- 1052
- 2080

(3 marks)

(iii) Assuming Least-Recently-Used (LRU) page replacement algorithm is used, identify three page references such as if they are the next three page references generated by the process, each will cause a page fault.

(3 marks)

(iv) For the page reference string identified in Q(b)(iii), will each page reference still cause a page fault if First-In-First-Out (FIFO) page replacement algorithm is used instead of LRU? What will be the resulting page/frame mapping in this case?

(4 marks)

(c) In a demand paging system, programs with good locality will generate less number of page faults.

(i) Describe the concept of locality.

(3 marks)

(ii) For the code fragments shown in Figures Q3c, assuming that array elements are stored according to rows and page size is 1,024 bytes, which fragment, the one on the left or the one on the right, will exhibit better locality? Explain your answer.

int array[1024][2014]; for (int i=0, i<1024, i++) for (int j=0; j<1024, j++) array[i][j]++;	int array[1024][2014]; for (int j=0, j<1024, j++) for (int i=0; i<1024, i++) array[i][j]++;

Figure Q3c

(3 marks)

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4. (a) (i) Compared to the tree directory structure, what is the advantage of acyclic graph directory structure?

(2 marks)

(ii) Protection bits of a file are set as "r w x r- x r - -" in a UNIX system. What are the permissions granted to the users who are in the owner's group?

(2 marks)

(iii) For each of the following two I/O scenarios, explain whether it would be more appropriate to use buffering or caching mechanism:

- Download a file from Internet and save it on hard-disk
- Read a file block repeatedly from hard-disk

(2 marks)

(iv) Disk block size may affect both performance and space utilization of a file system. Identify the problem that may arise when the block size is too small.

(2 marks)

(b) A system uses Unix-like inodes for file allocation. Each inode contains 10 direct, 1 single indirect, and 1 double indirect pointers. The block size is 512 bytes and the size of block pointer is 8 bytes. Suppose that the file position pointer is currently at 37,000 and the next 1,000 bytes are to be read from the file.

(i) What are the logical blocks that need to be accessed?

(2 marks)

(ii) Assuming that only the inode is in the memory, how many disk read operations need to be performed?

(4 marks)

(iii) What is the maximum file size that this inode structure can support?

(2 marks)

(c) Assume that a hard disk has 200 cylinders numbered from 0 to 199. At time t_0, the disk head is at cylinder 150 after servicing a request. Serving a request takes T + 20 units of time, where T is the number of cylinders to move. Table Q4a shows the arrival time (in time units) and destination cylinder number of seven requests.

Note: Question No. 4 continues on Page 8

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Table Q4a

Arrival Time	Cylinder Number
t_0+0	160
t_0+10	40
t_0+15	170
t_0+50	100
t_0+100	80
t_0+180	60
t_0+260	50

Give request servicing sequence for each of the following two disk scheduling algorithms:

- LOOK
- C-LOOK

It assumes that the disk head is currently moving in outwards direction (i.e., towards cylinder 199). For C-LOOK, it also assumes that requests are only serviced in the outwards direction.

(4 marks)

END OF PAPER

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Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**

2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.